

PS2: Heatwaves & Edge AI — Arduino UNO Q Reference

Problem Statement 2 (BPUT Hackathon 2026): Heatwaves and Edge AI, Using Technology to Improve Quality of Life with Arduino UNO Q. Local, room/street-level heat stress detection and response that keeps working through power and network outages.

Sensor Modules Used (This Project)

Parameter	Sensor Used	Notes
Temperature / Humidity / VOC	BME688	Digital — temp (°C), humidity (%RH), and VOC/gas index directly
Radiant Heat	DS18B20 (in black globe)	Digital -> °C; globe mounting approximates mean radiant temperature
Air Movement / Wind Direction	Anemometer with wind vane	Analog/pulse -> wind speed (m/s) and direction (°)
Light Intensity	BH1750	Digital (I2C) -> lux directly; proxy for solar load
Surface Temperature (roofs/walls)	MLX90640	Thermal array (I2C) -> 32x24 pixel temperature map, °C

System Architecture

- 3 nodes total: 1 master node + 2 slave nodes.
- Master node: Arduino UNO Q — runs Edge AI inference locally (Qualcomm Dragonwing QRB2210 Linux processor) with the STM32U585 handling low-power sensing and relay/actuation control, as mandated by the problem statement.
- Slave nodes: ESP32-based, each carrying the BME688 / DS18B20 / anemometer / BH1750 / MLX90640 sensor set, reporting readings back to the master.

Mandatory Hardware Constraint

Per the problem statement: all AI inference must run on the Arduino UNO Q itself — no cloud APIs or external AI services for core decision-making. Sensors, actuators, displays, and comms modules are open choice, but the UNO Q must remain the compute platform of the submitted prototype.

Core Outputs Expected

- Local computation of heat stress indices (Heat Index, WBGT) rather than raw temperature alone.
- Short-horizon prediction of how hot a specific room/space will get, learned from its own history.
- Risk grading personalised to the person/activity (labourer, child, elderly).
- Automatic local relief actions — fan/cooler, ventilation, evaporative cooling, or rest-break alerts.
- Continued operation through power cuts and network outages.
- Street-level heat map aggregated from multiple nodes for municipal planning.